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Ministry of Primary and Secondary Education — Republic of Zimbabwe

Heritage-Based Curriculum · School-Based Assessment · 50 Marks

SCHOOL-BASED PROJECT

NAME OF SCHOOL:	Mopani park secondary school
NAME OF PUPIL:	Praise Mutema
LEVEL / GRADE:	FORM 2
LEARNING AREA:	BUILDING STUDIES
DATE GENERATED:	19 May 2026

PROJECT TITLE

Accessible School Buildings and Facilities for All Students

Stage 1	Stage 2	Stage 3	Stage 4	Stage 5	Stage 6	TOTAL
5	10	10	10	10	5	50

SYLLABUS TOPICS

- Building Design and Planning
- Building Materials and Construction
- Safety and Security in Buildings
- Environmental Sustainability in Buildings

PROJECT OBJECTIVE

This project aims to improve accessibility in school buildings and facilities for students with disabilities at Mopani Park Secondary School. By applying Building Design and Planning concepts, we will design a more inclusive and accessible school environment. The project will also apply Building Materials and Construction knowledge to identify suitable materials for ramps and walkways. The outcome of this project will benefit all students, particularly those with disabilities, by providing equal access to learning resources and opportunities.

PROJECT DESCRIPTION

Mopani Park Secondary School has a diverse student population, including students with physical disabilities who face challenges navigating the school's buildings and facilities. The current school infrastructure presents several barriers, including uneven walkways, narrow corridors, and inaccessible toilets. This project seeks to address these challenges by designing and implementing accessible features in the school's buildings and facilities. The project will involve a thorough assessment of the school's infrastructure, identification of suitable building materials, and the design of ramps, walkways, and other accessible features.

STAGE 1: PROBLEM IDENTIFICATION (5 MARKS)

1.1 Description of the Problem

The school's main entrance at Mopani Park Secondary School is not wheelchair accessible, posing a significant barrier for students with mobility impairments. This problem affects students with physical disabilities, particularly those who use wheelchairs, and their caregivers. The school's main entrance is a long distance from the disabled toilet, forcing students to navigate through a narrow and uneven path, which is also slippery during the rainy season. This situation creates a challenge for students with mobility impairments to access the school's facilities, including the classrooms, library, and cafeteria. The lack of accessibility also affects the school's ability to provide inclusive education for all students.

1.2 Brief Statement of Intent

This project aims to investigate the accessibility of school buildings for students with disabilities at Mopani Park Secondary School, focusing on the main entrance and disabled toilet. The project will create practical solutions to improve the accessibility of the school's facilities, using Building Studies curriculum concepts such as the principles of universal design, accessibility standards, and building codes.

1.3 Project Specifications / Parameters

1. The project must design a wheelchair-accessible entrance for the school, incorporating features such as a ramp, wide doors, and a level entrance.
2. The project must ensure that the disabled toilet is easily accessible from the main entrance, with a smooth and even path.
3. The project must apply the principles of universal design to ensure that all school facilities, including classrooms, library, and cafeteria, are accessible to students with disabilities.
4. The project must comply with the Zimbabwean Building Code and accessibility standards for schools, as outlined in the Building Studies curriculum.

STAGE 2: INVESTIGATION OF RELATED IDEAS (10 MARKS)

Related Idea 1: Ramps for Easy Access

Evidence

Ramps are a common solution for providing easy access to buildings for people with mobility impairments. They work by using a gradual incline to connect two levels, allowing people to move from one level to another without having to climb stairs. This concept is grounded in Building Studies knowledge of slope and gradient, which is essential in designing accessible routes.

Merits

- Allows easy movement for people with mobility impairments
- Can be used in conjunction with lifts or elevators for added accessibility
- Can be designed to be visually appealing and blend in with the surrounding architecture

Demerits

- May require significant structural modifications to the building
- Can be expensive to install and maintain
- May not be suitable for all types of buildings or terrain

Related Idea 2: Adaptive Design for Wheelchair Users

Evidence

Adaptive design involves modifying existing buildings to make them more accessible for people with disabilities. This can include widening doorways, installing wheelchair-accessible toilets, and creating clear routes for easy navigation. Adaptive design is grounded in Building Studies knowledge of space planning and interior design.

Merits

- Can be a cost-effective solution compared to building new accessible facilities
- Allows existing buildings to be adapted to meet the needs of people with disabilities
- Can be designed to be flexible and adaptable to different user needs

Demerits

- May require significant modifications to the building's layout and design
- Can be challenging to implement in historic or culturally significant buildings
- May not be suitable for all types of buildings or user needs

STAGE 3: GENERATION OF IDEAS / POSSIBLE SOLUTIONS (10 MARKS)

Possible Solution 1: Accessible Ramps and Automatic Doors

Evidence

The solution involves designing and installing accessible ramps at all entrances of the school, ensuring that they are wide enough for wheelchairs to pass through comfortably. Automatic doors will also be installed to allow easy access for students with mobility impairments. This solution applies the Building Studies concept of accessibility in building design, which is essential for creating inclusive environments.

Merits

- Improves accessibility for students with mobility impairments
- Enhances the overall safety and security of the school environment
- Complies with the Zimbabwean Building Standards Act, which emphasizes accessibility in building design

Demerits

- High initial cost of installing ramps and automatic doors
- Potential for technical issues with automatic doors, such as jamming or malfunctioning
- May require significant modifications to the existing school infrastructure

Possible Solution 2: Adapted Classrooms with Assistive Technology

Evidence

The solution involves modifying existing classrooms to accommodate students with special needs, by installing assistive technology such as text-to-speech software, speech-generating devices, and communication boards. This solution applies the Building Studies concept of interior design and space planning, which involves creating functional and adaptable spaces that meet the needs of users. The assistive technology will be integrated into the classroom design to promote inclusivity and accessibility.

Merits

- Provides equal learning opportunities for students with special needs
- Enhances the overall learning experience for all students
- Can be easily integrated into the existing classroom layout and design

Demerits

- Requires significant investment in assistive technology and training for teachers
- May require modifications to the existing curriculum to accommodate students with special needs
- May require additional support staff to assist students with special needs

STAGE 4: DEVELOPMENT / REFINEMENT OF CHOSEN IDEA (10 MARKS)

4.1 Chosen Idea

Accessibility Ramps for Mopani Park Secondary School

4.2 Justification of Choice

1. The chosen solution was selected because it addresses the problem of inaccessible school buildings for the disabled students at Mopani Park Secondary School by providing a clear and direct path for them to access the school buildings.
2. This solution also aligns with the Building Studies syllabus topic of Design and Technology, specifically the concept of creating functional and safe spaces for users.

4.3 Developments / Refinements

Refinement 1: Inclined Plane Ramps with Handrails

The first improvement made to the accessibility ramps was the inclusion of inclined plane ramps with handrails to provide support and balance for students with mobility impairments. This refinement strengthens the solution by ensuring that students can safely navigate the ramps. Additionally, the handrails provide a secure grip for students who may need assistance.

Refinement 2: Tactile Pathway Markings

The second improvement made was the installation of tactile pathway markings to guide visually impaired students to the entrances of the school buildings. This refinement benefits the solution by providing a clear and accessible route for students with visual impairments, promoting independence and confidence.

Refinement 3: Regular Maintenance and Inspection Schedule

The third improvement made was the creation of a regular maintenance and inspection schedule to ensure that the accessibility ramps and tactile pathway markings remain in good condition. This refinement strengthens the solution by preventing potential hazards and ensuring that the accessibility features remain functional and safe for students.

4.4 Overall Presentation of Final Solution

The refinements to the accessibility ramps have resulted in a comprehensive and inclusive solution that addresses the needs of disabled students at Mopani Park Secondary School. The combination of inclined plane ramps with handrails, tactile pathway markings, and a regular maintenance and inspection schedule provides a safe and accessible environment for all students, promoting equality and dignity.

STAGE 5: PRESENTATION OF THE FINAL SOLUTION (10 MARKS)

Title of Presentation

"Accessible Pathways for the Disabled at Mopani Park Secondary School"

Type of Presentation

Service Presentation, with visual aids and a model of the accessible pathway, and an interactive presentation that demonstrates the use of ramps and gentle slopes.

Materials Used

- A large model of the Mopani Park Secondary School building
- A set of ramps and gentle slopes made of wood and steel
- A display board with diagrams and photographs of the original inaccessible pathways
- A set of PowerPoint slides with information about the Building Studies syllabus topics used in the project
- A short video demonstrating the use of accessible pathways in the school

Description of Final Presentation

The presentation includes a visual model of the accessible pathways, which demonstrates how the use of ramps and gentle slopes can improve accessibility for the disabled at Mopani Park Secondary School. The presentation also includes an interactive element, where the audience can walk through the model and experience the improved accessibility. The display board and PowerPoint slides provide information about the Building Studies syllabus topics used in the project, such as the importance of accessibility and the use of ramps and gentle slopes in building design. The short video demonstrates how the accessible pathways can be used in real-life situations.

STAGE 6: EVALUATION AND RECOMMENDATIONS (5 MARKS)

6.1 Relevance to Statement of Intent

Our project on school buildings and accessibility for the disabled at Mopani Park Secondary School successfully addressed the main issue of creating a more inclusive environment for all students. We achieved this by designing and proposing modifications to the existing school buildings to make them more accessible and safe for students with disabilities. The proposed solutions effectively met the requirements outlined in the Statement of Intent.

6.2 Challenges Encountered

- Insufficient budget for the proposed modifications
- Limited expertise in the school to undertake the modifications
- Resistance from some teachers and students to the idea of changing the existing school layout

6.3 Recommendations

1. The school administration should allocate a specific budget for the proposed modifications to ensure their timely completion.
2. The school should consider hiring a professional architect or contractor with expertise in accessible building design to oversee the modifications.
3. The school should establish a committee comprising teachers, students, and parents to promote awareness and support for the modifications and address any concerns or resistance.
4. The school should conduct regular assessments to monitor the effectiveness of the modifications and identify areas for further improvement.

